Sensor Report

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Abstract—A Formula Student (FS) driverless racecar depends on a high-rate, tightly synchronized sensor suite that acts as the car's eyes, ears and sense of balance. These sensors support perception (finding the track boundaries and cones), localization (knowing where the car is) and state estimation (tracking speed, yaw rate and slip). This report is a beginner-friendly, readable but in-depth guide to the sensors used on a driverless FS vehicle: LiDAR, stereo vision, GNSS/INS, inertial sensors, wheel-speed, Hall-effect and optical sensors, and the compute and actuation (steer-by-wire and brake-by-wire) they connect to. Each sensor's working principle, typical errors and fixes, how to integrate it on the car, approximate costs, and how to source it in India has been discussed. Report also includes current hardware, a recommended configuration for an Indian team, and testing and validation practice.

Keywords—autonomous racing, Formula Student Driverless, LiDAR, stereo camera, GNSS/INS, IMU, Hall-effect sensor, sensor fusion, steer-by-wire, brake-by-wire

I. Introduction

Sensors convert physical quantities, such as acceleration, angular velocity, distance, speed, position and light, into electrical or digital measurements that a computer can process. A driverless FS car has no human to see a cone or feel the car sliding, so the quality of these measurements directly decides how well the software can detect cones, rebuild the track, localize the car, plan a path and control the vehicle.

The sensors that matter most fall into five groups: Light Detection and Ranging (LiDAR) sensors, monocular and stereo cameras, Global Navigation Satellite System (GNSS) receivers, inertial measurement units (IMUs) and wheel-speed sensors. Depending on the architecture, steering-angle encoders, pedal and brake-pressure transducers, suspension-travel sensors and optical ground-speed sensors add information about the vehicle state and help monitor the health of the system.

Sensors feed perception and state estimation, which feed mapping, planning and control, which finally command the steering and brakes.

II. Sensor Fundamentals

A. Exteroceptive and Proprioceptive Sensors

Sensors are grouped by what they measure. *Proprioceptive* sensors measure the car itself and answer the question "How is the car moving?" Examples are IMUs (accelerometers and gyroscopes), wheel-speed encoders, steering-angle sensors and battery management systems. *Exteroceptive* sensors measure the world around the car and answer "What is around the car, and where?" Examples are LiDARs, cameras and radars. GNSS receivers too are considered here because they measure position against satellites outside the vehicle.

B. Key Specifications

Datasheets use a handful of terms again and again. Table I explains them in plain language, with examples we meet later in this report.

TABLE I. Key sensor specifications

Term	Meaning
Range	Minimum and maximum distance over which the sensor gives reliable data.
Resolution	Smallest change the sensor can tell apart, e.g., 0.1° between laser beams.
Field of view (FOV)	Angular span covered, horizontally and vertically, e.g., 360° × 30° for a spinning LiDAR.
Accuracy	How close a measurement is to the true value.
Precision	How tightly repeated measurements cluster. NOTE: A sensor can be precise without being accurate.
Update rate	How many data frames per second the sensor outputs.
Latency	Delay between a real-world event and its data reaching the computer.
Noise	Random, fast fluctuations on top of the true signal; reduced by filtering.
Bias	Constant offset, e.g., an IMU that reports acceleration while standing still.
Drift	Slow, accumulating error over time; critical for IMUs.

C. Active and Passive Sensing

Passive sensors, such as cameras, emit nothing and rely on ambient energy such as sunlight. They are light, low-power and hard to interfere with, but they struggle with glare, shadows, darkness, rain and fog.

Active sensors, such as LiDAR, radar and ultrasonic units, emit their own light, radio waves or sound and measure the reflection. They work independently of ambient light and measure distance directly, but they draw more power, weigh more, and can interfere with identical sensors nearby.

D. Interfaces and Protocols

The Controller Area Network (CAN) is a specialized, highly durable network protocol designed for vehicles. Instead of running heavy, complex bundles of individual wires from every single sensor to a central computer, CAN allows all electronic components to wire into a single, shared two-wire "highway" (bus). CAN Flexible Data-rate (CAN FD) is the modern upgrade to traditional CAN, designed to handle the data demands of semi-autonomous and fully autonomous vehicles.

TABLE II. COMMON SENSOR INTERFACES ON AN FS DRIVERLESS CAR

Link	Typical speed	Typical use
CAN	125 kbps to 500 kbps MAX: 1 Mbps	Driverless Status Indicator (ASB Lights, Tire and Brake Temperature Sensors
CAN FD	Slow phase (arbitration): 1 Mbps Fast phase (data): 2 to 5 Mbps with 8 Mbps under perfection conditions	Battery Management System, Inveter and Electric Motor controller, Advanced Driver Assistance Systems
Gigabit Ethernet	1 Gbit/s and above	3D LiDAR point clouds, machine-vision cameras.
USB 3.x	5 Gbit/s and above	Stereo cameras such as the ZED 2i. Use screw-lock cables.
UART / serial	9600 bit/s to a few Mbit/s	GNSS and INS configuration, NMEA text strings, microcontroller links.
PTP (IEEE 1588)	Negligible overhead on Ethernet	Sub-microsecond clock synchronization across networked sensors.

E. Calibration and Time Synchronization

1) Intrinsic calibration: Corrects the imperfections of the sensor itself. For a camera it models focal length, optical center and lens distortion so that pixels map to real-world rays; for a LiDAR it corrects the internal alignment of each laser beam. In essense, it matches pixel data to accurate physical coordinates.

2) Extrinsic calibration: Finds where each sensor sits, and how it is rotated, relative to the car's coordinate frame (usually the IMU or center of gravity), as a rotation matrix and a translation vector. Extrinsic calibration determines the exact physical position and orientation of a sensor relative to another sensor or the vehicle's central reference frame. *Translation Vector (T): The precise distance (x, y, z positions) from the sensor to the reference frame, usually measured in millimeters. Rotation Matrix (R): The precise angles (roll, pitch, yaw) showing how the sensor is tilted or turned relative to the vehicle's heading*

3) Even small errors, a few millimeters of translation or a fraction of a degree of rotation, shift projected LiDAR points noticeably at cone distances and corrupt cone detection.

4) Time synchronization: At speed, a tiny clock mismatch becomes a real distance. At 15 m/s (54 km/h), a 5 ms offset between a camera frame (at t = 100 ms) and a LiDAR sample (at t = 105 ms) equals 7.5 cm of travel, enough to misplace a cone. Hardware time synchronization, such as the Precision Time Protocol (PTP, IEEE 1588) or a GPS pulse-per-second (PPS) signal, anchors every timestamp to one master clock so that sensor fusion lines up.

III. PERCEPTION SENSORS

A. LiDAR

1) Working principle: A LiDAR fires a short laser pulse and measures the time Δt it takes to bounce off an object and return. Because the light travels out and back, the distance is half the speed of light multiplied by that time:

$$d = c \cdot \Delta t / 2 \quad (1)$$

where c is the speed of light. A LiDAR repeats this tens of thousands of times per second while sweeping the beam. Mechanical spinning units trace 360° circles, solid-state units steer the beam with MEMS mirrors or optical phased arrays, and flash LiDARs light the whole scene at once.

2) Types: Spinning (mechanical) LiDARs (e.g., Velodyne, Ouster, Hesai) rotate laser–detector pairs for full 360° coverage, which suits a central mounting spot on an FS car. *Solid-state* LiDARs have no large moving parts, so they tolerate vibration and shock better, but usually cover a narrower or differently shaped field of view. *2D* LiDARs sweep a single plane and are mainly used for simple wall-following or mapping, while *3D* LiDARs return multi-channel point clouds and are what FS Driverless normally uses to recognize cones of known height.

3) Channels, resolution and point clouds: The number of *channels* is the number of vertical laser–detector pairs, from 16 in budget units to 128 or more in high-end ones; more channels give a denser vertical profile. *Vertical resolution* is the angular gap between stacked beams, while *horizontal resolution* depends on rotation speed and firing rate. A cone 20 m away is small, so the sensor needs enough resolution for the cone to receive several returns. The output is a *point cloud*: a set of (X, Y, Z) coordinates, each with a return intensity and a timestamp.

4) Cone detection pipeline: First, *ground removal* filters out the track surface using methods such as RANSAC plane fitting, Patchwork or grid-based approaches. Then *clustering* (e.g., Euclidean cluster extraction or DBSCAN) groups the remaining points into objects, and each cluster is checked for height, width and shape to decide whether it is a cone. Color comes from the camera.

5) Limitations:

a) Direct sunlight adds near-infrared light that can saturate the detectors and reduce range.

b) Dust kicked up by cars and heavy rain can cause backscatter, producing "ghost points".

c) Glossy carbon fiber, wet tarmac and retro-reflective stickers can scatter laser energy away from the receiver.

d) LiDARs also cost much more than cameras, although cheaper units have made them reachable for student teams.

6) LiDAR: Hesai Pandar40P: The Pandar40P is a 40-channel mechanical 3D LiDAR with a 360° horizontal field of view, a range of up to roughly 200 m on reflective targets, and an Ethernet interface. On our car it provides the main point cloud for cone detection. A cone-marked track is short enough that range is rarely the limit; resolution on distant cones and mounting height matter more. Hesai units can take time from a GPS PPS signal or PTP, so they can share a clock with the other sensors. Check the current datasheet and availability before buying, because Hesai has released newer models.

B. Cameras

1) Types of cameras:

a) Monocular cameras give a single 2D image stream; they are light, cheap and need little computing, but must infer depth.

b) Stereo cameras use two synchronized lenses a fixed baseline apart – it mimics human binocular vision. Because your two eyes are separated by a small distance, objects look slightly shifted between your left and right eye; similarly matching pixels between the two images (the disparity) gives depth directly.

c) RGB-D camera is a sensor that captures both color video and dense, per-pixel depth information simultaneously**.** This camera adds an active infrared depth sensor, which works well indoors but poorly in bright sunlight.

d) Event cameras (also called a neuromorphic camera – they are called so because the hardware is intentionally designed to mimic the biological structure and behaviour of the human nervous system and retina) does not take photos or videos; Essentially it captures light by sending a continuous stream of "digital coordinate notes" every time a single pixel senses a change while reportking only per-pixel brightness changes, offering microsecond timing, high dynamic range and almost no motion blur, but they need specialized algorithms.

2) Shutter, resolution and dynamic range:

A global shutter acts like a studio flash, freezing the entire image instantly to eliminate motion distortion, while a rolling shutter records row-by-row like a scanner. This choice, along with standard 720p to 1080p resolutions and HDR capabilities, ensures autonomous vehicles receive sharp, accurately positioned images even at high speeds or under extreme lighting shifts. In autonomous driving, a rolling shutter is dangerous because a warped image can trick the AI into miscalculating the exact position or shape of a traffic sign, pedestrian, or lane line.

While consumer smartphones use massive 4K or 8K resolutions, autonomous vehicles deliberately stick to 720p or 1080p. High resolutions generate massive data streams that cause computer lag (latency). Lower resolutions, running at high frame rates (30 to 120 frames per second), keep the data stream light and lightning-fast so the car can react instantly.

When a car drives under a dark bridge on a blazing sunny day, a standard camera gets blinded, rendering the tunnel pitch black and the exit completely washed out. An HDR camera blends multiple exposures simultaneously, allowing the car's computer to see hidden details in both the deepest shadows and the brightest glare at the exact same time.

3) Optics: A short focal length acts like a wide-angle lens to capture a broad view (e.g., 120°) of the track edges, while a long focal length acts like a zoom lens to focus sharply on obstacles far down the straightaway. Because wide lenses heavily warp the image, software must first "un-distort" (rectify) the curved lines back into straight lines so the AI can accurately measure distances.

4) Cone detection and color: Autonomous vehicles use two different methods to spot track cones: a highly accurate deep-learning AI system (YOLO) as the primary driver, and a simpler, math-based color filter (HSV/LAB) as a lightweight backup to cross-check the results. This dual-layer approach ensures that if the heavy AI system lags or gets confused by lighting, the vehicle still has a fast, bulletproof safety net to keep it on the track.

a) YOLO (You Only Look Once) is a state-of-the-art AI model famed for its incredible speed and accuracy. When fed a camera frame, it looks at the entire image just once and instantly handles two critical tasks:

- Bounding Boxes: It draws a tight digital box around every cone to pinpoint its exact pixel location.
- Color Classification: It instantly categorizes the cone into its specific Formula Student (FS) class—blue, yellow, small orange, or big orange—which tells the car's navigation system where the left boundary, right boundary, or special zones (like the start/finish line) are.

b) HSV/LAB Filtering Instead of using a complex digital "brain," the classical pipeline uses basic pixel math. It isolates specific colors by converting the image into specialized color spaces:

- HSV (Hue, Saturation, Value): Separates color type (hue) from how vibrant or bright it is.
- LAB Space: Separates lightness from color dimensions (A for green-to-red, B for blue-to-yellow).

5) Depth estimation: With a stereo pair, depth Z is inversely proportional to the disparity d:

$$Z = f \cdot B / d \tag{2}$$

where f is the focal length in pixels and B is the baseline between the lenses. A cone twice as far away produces half the disparity, so depth accuracy falls with distance. With a single camera, distance can be estimated from the known height of an FS cone and its height in pixels.

6) Limitations: Low sun and glare can saturate pixels, long exposures cause motion blur that smears edges, and quick changes between sun and shadow make auto-exposure lag, so cones blend into the track.

7) The ZED 2i: Stereolabs ZED stereo cameras (specifically the ZED 2 and ZED 2i) are the gold standard for depth sensing in Formula Student Driverless (FSD) competitions globally. It is a *stereo* camera that can also be used as a monocular camera. The ZED 2i provides an integrated, ruggedised perception solution tailored for harsh racing environments, combining structural durability with built-in spatial sensors

a) IP66 Ruggedised Enclosure: Features a dust-tight, water-resistant aluminum frame that replaces the older plastic chassis. This design isolates internal components from severe chassis vibrations, track debris, and sudden weather changes typical of outdoor racing.

b) Neural Depth Sensing: Employs dual synchronised lenses to enable stereoscopic vision. The integrated software utilizes neural networks to generate real-time depth maps, enabling accurate distance measurement to track-boundary cones at ranges up to 20–35 metres.

c) Integrated Polarization: Utilises an internal circular polarizing (CPL) filter layer. This hardware feature mitigates solar glare from the asphalt and vehicle bodywork, preventing image overexposure and preserving edge-detection data for the vision pipeline.

d) Sensor Fusion Capabilities: Integrates an internal 9-Degrees of Freedom IMU, gyroscope, and barometer. These high-rate inertial sensors stream spatial data directly into Robot Operating System (ROS) nodes, allowing engineers to fuse visual data with wheel encoders and LiDAR to calculate vehicle slip angles and path trajectorie

C. Radar and Other Perception Sensors

Radar, ultrasonic, and thermal sensors are generally omitted from Formula Student (FS) vehicles because their underlying physics are fundamentally incompatible with small, plastic racing cones. While effective for standard passenger cars, their limitations under racing conditions make them impractical.

a) Radar

- Mechanism: Emits radio waves and measures the Doppler shift (the change in the frequency of a returned radar wave caused by relative motion between the radar system and a target) to calculate an object's relative velocity instantly.
- FS Limitation: Radio waves require dense or metallic surfaces to bounce back. Small, low-profile, soft plastic cones absorb or scatter the signal, making reliable detection impossible.

b) Ultrasonic Sensors

- Mechanism: Utilizes micro-sonar sound pulses to measure close-range proximity based on echo timing.
- FS Limitation: Restricted to a range of only a few meters and suffers from slow refresh rates. At racing speeds, the lag between audio pulses results in dangerously outdated spatial data.

c) Thermal Cameras

- Mechanism: Detects infrared radiation (heat signatures) instead of visible light, bypassing nighttime visibility constraints.
- FS Limitation: Relies entirely on temperature contrast. Cones quickly equalize to the same temperature as the asphalt, rendering them invisible on a thermal feed. High hardware costs offer no practical return on investment.

IV. Localization and State-Estimation Sensors

A. GNSS

1) Basics: A Global Navigation Satellite System (GNSS) receiver listens to timed radio signals from satellites in orbit (GPS, GLONASS, Galileo, BeiDou). By measuring how long each signal took to arrive, it computes distances ("pseudoranges") to at least four satellites and solves for latitude, longitude, altitude and its own clock error.

2) Standard GNSS, RTK and PPK: Satellites are thousands of miles away, hence as signals pass through Earth's atmosphere they get delayed and distorted. In fact, standalone GNSS typically has a horizontal error of 2 to 5 m, too coarse for racing. In order to tackle this problem, engineers introduced a secondary component: a base station.

a) The Real-Time Kinematic (RTK) uses a fixed base station (usually a tripod next to the track) at a known location that sends correction data by radio to the receiver on the car, reaching centimeter-level accuracy (about 1 to 2 cm).

b) Post-Processed Kinematic (PPK) logs raw data on both the car and the base and corrects it after the run. Radios can drop connections or suffer from interference on a racetrack. Since PPK doesn't rely on a live radio link, you never lose data. This makes it perfect for Track Mapping (driving the track slowly before the race to create a perfect map) or establishing a Ground Truth (a flawless reference line used by engineers to check if their other sensors were lying during a test run).

3) Dual-antenna heading: A single-antenna GNSS receiver can derive a vehicle's heading vector only when there is sufficient forward velocity to calculate the change in position over time. At low speeds or when completely stationary, single-antenna velocity vectors break down, leaving the system blind to the vehicle's true orientation. To resolve this, a dual-antenna configuration is deployed along a fixed baseline parallel to the vehicle's longitudinal axis. By measuring the carrier-phase difference of the incoming satellite signals between the primary and secondary antennas, the system performs integer ambiguity resolution to compute an instantaneous, high-precision heading (yaw) angle. This phase-interferometry method functions completely independently of a magnetometer—rendering it immune to magnetic interference from high-voltage EV powertrains—and guarantees instantaneous heading initialization at system startup before the vehicle even moves.

4) Limitations: Multipath happens when signals bounce off barriers, metal stands or bodywork and produce position jumps. Obstructions such as bridges, pit buildings and trees block satellites and can drop an RTK fix (carrier ambiguities). Receivers also update at only about 5 to 20 Hz, far slower than a control loop, so GNSS must be fused with a faster IMU and wheel speeds through a Kalman filter.

B. Inertial Measurement Units

1) What an IMU measures: An IMU is a single sensor package that combines multiple individual components. An *accelerometer* measures proper acceleration along three axes, including gravity, in m/s². On an Formula car it 'senses' launch, cornering and braking. A *gyroscope* measures rotation rate around the roll (tilt side-to-side), pitch(tilt front-to-back) and yaw(turning left-or-right) axes. The yaw rate (speed at which the vehicle changes its heading) is critical for

cornering. A *magnetometer* measures the magnetic field to find heading relative to magnetic north. It is important to note that on an electric FS car the high-voltage battery, inverter and motor create interference that often makes it unreliable or unusable. *NOTE: Strictly speaking, a traditional IMU consists only of Gyroscopes (measuring rotation rate) and Accelerometers (measuring linear acceleration and gravity) — giving you 6 Degrees of Freedom (DoF). If a Magnetometer is added to the package, it is technically called a MARG sensor (Magnetic, Angular Rate, and Gravity), though many manufacturers and engineers loosely call a 9-DoF package an IMU anyway.*

2) MEMS, FOG and RLG: Instead of spinning wheels, MEMS (Micro-Electro-Mechanical Systems) gyros use microscopically small silicon structures etched onto a microchip. These structures are continuously forced to vibrate. When the car turns, the Coriolis effect pushes against these vibrating pieces, causing a tiny measurable shift in their movement. Moreover, since they are etched directly onto silicon wafers (just like computer processors), they can be mass-produced at cheaper rates. They have no large moving parts to snap during a crash or heavy vibration, making them incredibly tough, light, and cheap.

Fiber-Optic Gyros (FOGs) use the **Sagnac effect**. A beam of laser light is split into two and sent in opposite directions through a long, tightly wound coil of optical fiber. If the car is perfectly still, both light beams exit the coil at the exact same time. If the car rotates, one beam has a slightly shorter path to travel than the other. The sensor measures the phase shift (interference) between the two beams when they exit. To get high accuracy, you need hundreds of meters of fiber optic cable wound into a coil. This coil, along with the necessary laser diodes and shielding, makes the sensor heavy, bulky, and expensive to manufacture.

Ring laser gyros (RLG) operate by splitting a laser into two counter-propagating beams inside a sealed, rigid cavity and give navigation-grade accuracy, but are too large and costly for a student car.

3) Bias, drift and error growth:

a) Bias: A constant, predictable offset added to every measurement, caused by manufacturing imperfections or temperature changes. In the case, if the car is driving straight, the sensor constantly reads **+2°** instead of 0°.

b) Noise: Random, high-frequency jitter that averages out over long periods but degrades short-term accuracy. An example could be the sensor reads +0.5°, then -0.3°, then +0.1°, then -0.4° rapidly every millisecond.

c) Drift: A slow, time-dependent wandering of the sensor's average output, typically caused by thermal fluctuations. If the car calculates its position by adding up its steering inputs over time, a **+2° bias** means that after 1 second the error is 2°, after 5 seconds it is 10°, and after 30 seconds it is 60°.

$$\Delta v = b \cdot t, \quad \Delta x = \tfrac{1}{2} \cdot b \cdot t^2 \qquad (3)$$

For example, a bias of only 0.01 m/s² gives a 0.5 m position error after 10 s. Gyroscope bias makes the heading error grow with time as well, so a standalone IMU diverges within seconds without help from GNSS or wheel speeds.

C. Accelerometer and Magnetometer Technologies

IMU datasheets hide a choice of sensing technology. Table III compares the main types so that a datasheet can be read with confidence.

TABLE III. ACCELEROMETER AND MAGNETOMETER TECHNOLOGIES

Type	How it works	Notes for FS
Capacitive MEMS accelerometer	A tiny proof mass moves between plates; the capacitance change gives acceleration.	Used in most IMUs; cheap, small, responds to steady acceleration; limited at very high g and vibration.
Piezoresistive accelerometer	Strain in a silicon beam changes its electrical resistance.	Tolerates high shock; common in rugged units; temperature sensitive.
Piezoelectric accelerometer	A crystal generates charge when squeezed by the mass.	Excellent for vibration analysis; cannot measure steady (DC) acceleration.
Servo (force-balance) accelerometer	A feedback coil keeps the mass centered; coil current gives acceleration.	Very accurate and stable; larger and costlier; found in high-grade INS.
Hall-effect magnetometer	A voltage appears across a semiconductor in a magnetic field.	Cheap and robust but less sensitive; mainly for position and current sensing.
Magnetoresistive (AMR/GMR/TMR)	A thin film's resistance changes with the field.	Common in IMU compasses; disturbed by nearby magnets and steel.
Fluxgate magnetometer	A magnetic core is driven into saturation and the field imbalance is measured.	Very accurate and low-noise; larger and costlier.

Whatever the technology, a magnetometer on an electric car must be mounted away from motors, inverters and thick current-carrying cables and calibrated in place (hard-iron and soft-iron calibration corrects for steel and magnets on the car), or replaced by dual-antenna GNSS heading.

D. Inertial Navigation Systems and the VectorNav VN-200

An *inertial navigation system (INS)* combines a high-rate IMU with a GNSS receiver and runs a Kalman filter inside the unit (essentially this is sensor fusion and Kalman filter works best here as these have opposite strengths and weaknesses) GNSS gives drift-free global position at a low rate and can drop out; the IMU gives smooth, high-rate motion data that bridges the gaps. Together they output a continuous stream of position, velocity and orientation suitable for path tracking.

The VectorNav VN-200, is a GNSS-aided INS. It contains (an IMU) a three-axis accelerometer, gyroscope and magnetometer, a barometric pressure sensor and a GNSS

receiver, and runs an onboard Kalman filter that outputs at up to 200 Hz.

Two practical points matter for FS. First, it is a single-antenna, standard-GNSS device, so its absolute position is accurate to meters, not centimeters, unless external correction is added, for example from an RTK receiver. That is fine for relative motion over a lap but not for surveying a track. Second, its magnetometer should be used with care, or disabled, on an electric car (Section IV-C). Pairing it with an RTK GNSS module, as proposed in Section XI, closes this gap.

V. Wheel, Drivetrain and Vehicle-Dynamics Sensing

A. Wheel-Speed Sensors

1) Hall-effect sensors: These are active sensors that need power. A semiconductor detects the changing magnetic field of a toothed (tone) ring or of magnets as they rotate and outputs a clean digital square wave. They work down to zero speed, so they suit launch and low-speed control.

2) Inductive sensors: Passive coils wound around a magnet. A passing iron tooth changes the magnetic flux and induces an AC voltage. They are rugged, but the signal is too weak at very low speeds, so the output drops out.

3) Optical encoders: A slotted disk interrupts the beam between an infrared LED and a photodetector. They give very high resolution but are vulnerable to brake dust, moisture and debris.

B. The Hall Effect in Plain Language

When current flows through a thin conductor or semiconductor and a magnetic field passes through it at right angles, a small voltage appears across its sides. This is the *Hall effect*, and the voltage is proportional to the field strength.

A raw Hall voltage is microscopic. A Hall sensor chip takes that tiny electrical signal, amplifies it internally, and turns it into a clean, usable digital or analog signal that the car's computer can read easily. On the car the same principle appears in several places: wheel-speed pickups, non-contact pedal and steering-position sensors, brushless-motor commutation, and current sensing in the powertrain. Its strengths are no moving parts, no contact wear and tolerance to dirt and moisture. Its main weakness is sensitivity to stray magnetic fields, such as those near motors and magnets.

C. Optical Sensors

Optical sensors turn light into an electrical signal using photodiodes or phototransistors, usually paired with an LED or a laser. On the car they appear in several forms: optical encoders (above), reflective and slotted interrupter sensors, camera image sensors (Section III-B), LiDAR detectors (Section III-A) and optical ground-speed sensors. The last of these look down at the track surface and measure how its texture moves past (or use laser Doppler principles) to find true longitudinal and lateral velocity without relying on the wheels. They give an independent reference for tire slip angle, but they are expensive and sensitive to dirt, spray and mounting height, so on a student budget they are mostly used for validation rather than as the main speed source.

D. Motor Encoders and Resolvers

In an electric vehicle, optical or magnetic encoders mounted on the motor shaft precisely track rotor position and speed, streaming this data to the inverter to execute **field-oriented control (FOC)**—the mathematical algorithm that dynamically aligns the motor's magnetic fields for maximum torque and efficiency.

While encoders offer very high resolution, their delicate optical disks or internal electronics can struggle in harsh environments. To handle extreme conditions, a **resolver** acts as a rugged rotating transformer: an AC reference signal applied to the rotor induces sine and cosine voltage signals in the surrounding stator coils, encoding the exact absolute angle immediately upon startup without requiring calibration.

Because they lack fragile glass disks, LEDs, or microchips, resolvers easily tolerate high heat, severe vibrations, and heavy electromagnetic interference (EMI), making them the preferred choice for high-stress motorsport and heavy-duty EV powertrains

E. Odometry and Wheel Slip

Odometry estimates how far the car has moved by multiplying wheel angular speed by the tire's effective rolling radius and combining the result with the steering angle. It assumes the tires roll without slipping. In hard acceleration, braking and cornering that is not true, so wheel odometry drifts quickly and must be fused with the IMU and GNSS.

F. Steering, Pedal, Brake-Pressure and Suspension Sensors

1) Steering angle: Measured with optical encoders, magnetic (Hall) sensors or potentiometers on the steering shaft or at the rack and tie-rod. In autonomous operation it is the feedback for the steering controller and the lateral vehicle model.

2) Pedal position and brake pressure: Dual-redundant potentiometers or non-contact Hall sensors measure pedal travel. Piezoresistive or strain-gauge pressure transducers in the brake lines measure actual braking force, independent of linkage flex. These feed telemetry, regenerative-braking control and the plausibility checks required by the rules, such as the brake system plausibility device (BSPD)

3) Suspension travel: Linear or string potentiometers, or linear variable differential transformers (LVDTs), across the dampers measure ride height, roll, pitch and heave. They are used to tune springs and dampers and to understand weight transfer.

VI. Actuation: Steer-by-Wire and Brake-by-Wire

Sensors only matter if the car can act on what they report. In a driverless FS car the autonomous system must be able to steer, brake and request torque without a human. *Steer-by-wire* and *brake-by-wire* replace or supplement the mechanical driver link with an electrically commanded actuator, and each depends on sensors for feedback.

A. Steering Actuation

A steering actuator is usually a brushed or brushless motor with a gearbox or belt connected to the steering column or rack, controlled in position using a steering-angle sensor. The

key design numbers are: required torque, speed (the time to move from lock to lock) and holding behavior at standstill.

Redundant position sensors on the output are recommended so that a single sensor fault can be detected. Typically, the actuator's own controller runs the fast inner position loop, while the autonomous computer sends a target angle over CAN.

The system must also fail safe: if power or communication is lost, the steering must not command a sudden, large movement.

B. Brake Actuation and the Emergency Brake System

Formula Student driverless rules require two separate braking systems: an active brake-by-wire system for racing manoeuvres, and a fail-safe Emergency Brake System (EBS) managed by hardwired logic.

a) Service Brake: Brake-by-Wire

- Actuates the master cylinder or brake pedal using a linear actuator or an electro-hydraulic unit.
- Employs brake-pressure sensors to close the control loop. This allows the vehicle's central computer to request and maintain precise deceleration rates or hydraulic pressures dynamically.

b) Emergency Brake System (EBS): Fail-Safe Architecture

- Built using spring-applied brakes held open by continuous pneumatic pressure (pressure exerted by air or gas). If the vehicle loses electrical power or pneumatic pressure, the springs instantly release and mechanically clamp the brakes.
- Triggered by hardwired safety logic rather than software routines. This guarantees activation even during a complete autonomous computer lockup or operating system failure.
- Integrates tank-pressure transducers (sensor that measures the gas pressure inside a pneumatic) to continuously feed structural health data to the safety system, confirming that the pneumatic circuit is pressurized and the EBS is armed.

C. Feedback Sensors, Redundancy and Latency

For both actuators, sensor feedback closes the loop: steering angle, brake pressure, motor speed and torque, and actuator current. Their update rate and latency are part of the pipeline budget in Section VIII.

Redundant or cross-checked sensors let the safety system detect a stuck or failed reading, and the response to a failed check should be a defined safe state, such as a controlled stop or the EBS (Section IX).

VII. Sensor Fusion and Algorithms

A. Why Fuse Sensors

No sensor is perfect. Cameras give color but weak depth in poor light, LiDAR gives accurate 3D shape but no color, IMUs are fast but drift, and GNSS is absolute but slow. Fusion lets each sensor cover the weaknesses of the others and also provides redundancy: if one sensor is blinded or fails, the others can keep the car safe.

B. Kalman Filters

Imagine driving blindfolded while estimating your position from speed and steering (the *prediction*), and every so often peeking at a map (the *update*). Both the estimate and the peek are uncertain.

A Kalman filter is a mathematical algorithm that estimates the true state of a system over time by combining a series of noisy sensor measurements with a dynamic prediction model. The standard *Kalman filter* suits linear systems.

The *Extended Kalman Filter (EKF)*, the most common choice on autonomous racecars, it takes the messy, nonlinear equations of a car moving in 2D/3D space and uses calculus (Jacobian matrices) to draw a straight "tangent line" approximation at the current estimate. It is computationally cheap and lightning-fast.

The *Unscented Kalman Filter (UKF)* instead passes a small set of carefully chosen sample points (sigma points) through the nonlinear equations, which can be more accurate without needing derivatives, however is reserved for more complex systems as it requires more computational power.

C. Particle Filters

a) The Core Idea: A Cloud of Hypotheses

Imagine your FSD car is dropped onto a track with a blindfold on—its starting position is completely unknown. This is known as global localization from absolute scratch.

- Instead of guessing just one location, the algorithm scatters a "cloud" of individual particles all over the digital track map.
- Each particle acts as a distinct hypothesis representing a possible state (position and orientation) of the vehicle. Think of them as a swarm of invisible "ghost cars."

b) The Technical Loop: Measure, Compare, and Cull

As the real car moves, the algorithm runs a continuous "survival of the fittest" loop to narrow down where the car actually is:

- Sensor Comparison: The system compares actual sensor readings (such as LiDAR or camera cone data) against what each individual particle *should* see if it were really standing at that hypothetical location.
- Weighting: Particles that align closely with the real-world sensor measurements are assigned high probability weights.
- Resampling (Survival & Multiplication): The matching particles survive and multiply (replicating themselves in the cloud), while the particles whose guesses contradict the sensor data are heavily penalized and discarded (die out).

c) Why It Matters for FSD

- Finding Your Way From Scratch: Because of this aggressive filtering process, the chaos of the initial particle cloud rapidly collapses. Within just a few seconds of driving, all surviving particles cluster tightly around the *one true location* of the car.
- The Engineering Trade-off: While a Kalman filter is incredible at *tracking* motion once you already know where you are, a particle filter is uniquely powerful

for solving the "kidnapped robot" problem (finding yourself when completely lost). The catch is that managing thousands of particles takes massive computational power, which is why FSD cars often use it strictly for initial global localization at startup before handing off to a fast, lightweight Kalman filter for the actual race.

A *particle filter*—or *Monte Carlo Localization*—estimates a vehicle's position and orientation (*state*) by maintaining a "cloud" of individual *particles*, where each particle acts as a distinct hypothesis of where the vehicle might be.

It answers the question of how an autonomous vehicle handles global uncertainty from a complete cold start.

To tie that directly back to a Formula Student Driverless car, here is what is happening under the hood when those particles evaluate sensor data:

- Simulated Scans (Ray Casting): For every single "ghost car" particle, the algorithm uses the track's pre-mapped cone coordinates to calculate what a LiDAR or camera *should* see from that exact hypothetical position.
- Weighting and Resampling: If a particle's simulated cone layout matches the car's actual physical sensor feed, it gets a high weight; if it sees a completely different part of the track, its weight drops and it gets filtered out during resampling.

Because particle filters can handle complex, multi-modal distributions (unlike a standard Kalman filter which assumes a single neat bell curve), they are the go-to tool for finding out where the car is on the track map when it first boots up.

As the vehicle moves, the algorithm compares actual sensor readings (such as LiDAR or camera data) against what each particle *should* see from its hypothetical location: particles that align closely with real measurements are assigned high probability weights and replicated (*survive and multiply*), while those that contradict sensor data are discarded (*die out*). This probabilistic process makes particle filters uniquely effective for *global localization*, allowing a vehicle to reliably determine its true position from scratch even when its initial location is completely unknown. particle filter represents the possible states of the vehicle by a cloud of particles. Particles that agree with the measurements survive and multiply, and the others die out. They are useful for global localization when the starting position is unknown.

D. LiDAR–Camera Fusion for Cones

Using the intrinsic and extrinsic calibration, 3D LiDAR points are projected onto the camera image. When a LiDAR cluster falls inside a neural-network bounding box, the cone's color is assigned to the 3D cone position. This combines the accurate position from the LiDAR with the color from the camera.

E. SLAM

Simultaneous Localization and Mapping (SLAM) allows an autonomous car to build a map of an unknown track while simultaneously tracking its own position within it—a requirement for the Trackdrive event, where the track layout is completely unknown at the start. FastSLAM accomplishes this by pairing particle filters with individual landmark estimates for each particle. Alternatively, Graph-based SLAM models car poses and track cones as nodes in a network linked by measurement constraints, optimizing the full graph to maintain accuracy; due to its robustness, it is widely used in Formula Student driverless stacks. Finally, Loop Closure detects when the car returns to a previously visited location (such as the start line) and uses that landmark match to eliminate accumulated sensor drift across the map.

F. Coordinate Frames and Transforms

Every sensor has its own coordinate frame: the camera looks along its optical axis, the LiDAR is centered on its bracket, and the IMU follows its chip. A transform tree (the ROS 2 TF library) publishes the translation and rotation between all sensor frames, the vehicle frame and the map frame, so that every measurement is converted into one common frame before fusion and control.

VIII. COMPUTE, SOFTWARE AND DATA

A. Onboard Compute: NVIDIA Jetson AGX Orin

The AGX Orin is an all-in-one chip (System-on-Module) designed specifically for mobile AI applications like autonomous driving.

- *The Hardware Architecture*: It integrates a multi-core ARM CPU (to handle general code like vehicle steering logic) alongside a CUDA cores (handle traditional pixel shading, graphics rendering, physical stimulations and basic arithmetic) and Tensor Cores. The Tensor Cores are specialized hardware accelerators designed strictly to speed up matrix math, allowing the car to run heavy neural networks (like YOLO cone detection) and process 3D LiDAR point clouds simultaneously.
- *Power Efficiency*: Its biggest advantage is configurable power modes. Engineers can digitally throttle the chip to run at specific power profiles (e.g., capping it at 15W, 30W, or 50W). This gives teams the exact control needed to maximize processing speeds without exceeding the race car's thermal limits or draining its low-voltage battery.

Jetson Orin NX / Nano

Scaled-down, low-cost versions of the AGX module designed for strict space and budget limits. While highly cost-effective and draws significantly less electrical current, it features fewer CUDA and Tensor cores, reducing computational headroom. Moreover, adding multiple sensors can cause 100% hardware utilization and dangerous processing lag (latency).

Ruggedised Industrial PCs

High-end desktop architectures protected by vibration-proof shells for heavy computational loads. While it pairs x86 CPUs with mobile NVIDIA RTX graphics to run massive, complex code pipelines effortlessly. It introduces extreme weight and power penalties, drawing hundreds of watts and requiring bulky cooling systems.

B. Middleware and Frameworks

Most FS driverless stacks run on ROS 2, which provides (topics) publish–subscribe communication between software nodes, tools such as TF (transformation) and rosbag, and a large ecosystem (Nav2 – the official navigation and localization framework, PCL – massive standalone open-

source library, OpenCV – industry standard for 2D image processing). Teams usually add their own state-estimation, SLAM and planning nodes for the Acceleration, Skidpad, Autocross and Trackdrive events.

C. Logging, Playback and Simulation

Rosbag records raw sensor streams (point clouds, images, IMU, CAN) to fast NVMe storage during track runs, for offline debugging, tuning and repeatable replay. *Simulators* such as FSSIM or FS driverless simulators built on Gazebo or Unreal Engine model vehicle dynamics, tires and cone layouts, so teams can train and test software safely before running on the car.

D. Latency Budget

Latency is the delay from a real-world event, such as a cone entering the camera view, to the actuator command. At 15 m/s every 10 ms of delay equals 15 cm of travel, so the whole pipeline should be kept to roughly 50 to 100 ms. Table IV shows a typical budget. Keeping latency low prevents phase lag in the control loop, so the car does not overshoot the track boundaries.

TABLE IV. TYPICAL LATENCY BUDGET (APPROXIMATE)

Stage	Time
Sensor exposure and capture	5–10 ms
Data transmission (Ethernet, USB, CAN)	2–5 ms
Perception (YOLO inference, LiDAR clustering)	15–30 ms
State estimation and SLAM (EKF update)	5–10 ms
Path planning and control	5–10 ms
Actuator response	5–10 ms

IX. SAFETY AND COMPLIANCE

The points below summarize the Formula Student driverless rules as we understand them. Always check the current rulebook of the event you enter, because details change between years.

A. Rules That Touch the Sensors

1) Autonomous system (AS): The driverless system must operate without driver input. The sensors form the perception loop that detects the track limits and supports driving decisions.

2) Emergency brake system (EBS): An independent system that brings the car to a safe stop. It is triggered when safety loops fail, sensor communication drops or software heartbeats are missed.

3) Autonomous system status indicator (ASSI): Colored lights that show the autonomous state to marshals and scrutineers.

- *AS Off (Dark): System is unpowered or initializing.*
- *AS Ready (Continuous Yellow): Self-checks passed and awaiting the start signal.*
- *AS Driving (Flashing Yellow): Vehicle is operating autonomously at a 2–5 Hz flash rate.*
- *AS Finished (Continuous Blue): Mission is complete and the vehicle has safely stopped.*
- *AS Emergency (Flashing Blue with Sound): Fault detected or Emergency Brake System triggered, paired with an intermittent acoustic warning*

4) Remote emergency system (RES): A radio-controlled kill switch operated by marshals. It cuts the drive and triggers the EBS independently of the onboard sensors and computers.

B. Redundancy and Health Monitoring

One failed sensor at speed can cause a crash, so teams cross-check sensors, for example IMU dead reckoning (process of calculating your current position, velocity, and orientation over time by continuously measuring data using an IMU) against wheel odometry and GNSS, so that losing one stream does not blind the localization. Every sensor driver and perception node also runs under a software *watchdog timer*: if a LiDAR or camera stops publishing, or falls below its minimum rate (for example, no frame for over 50 ms), the watchdog raises a fault.

C. Failure Modes and Fallbacks

When a non-critical sensor fails, the stack degrades gracefully, for example by relying on the LiDAR if the camera is blinded by glare. If a safety-critical sensor or core node fails, or if localization diverges beyond a safe limit, the stack commands a controlled stop or triggers the EBS.

X. INTEGRATION AND TROUBLESHOOTING

A. Mounting and Vibration

LiDARs and stereo cameras need stiff brackets (carbon-fiber composite or machined aluminum such as 6061-T6). Even small flex adds noise and changes the extrinsic calibration, which corrupts point-cloud projection and SLAM. Isolation mounts can protect optics and delicate electronics from shock and fatigue, but any isolator changes how a sensor moves relative to the chassis, so re-run the extrinsic calibration after mounting and keep the IMU's orientation fixed.

B. Aerodynamics and Placement

Exterior sensors such as roof LiDAR pods and forward camera housings disturb the airflow, add drag and can shift the front-to-rear downforce balance. Sensors must sit high enough for a clear view over the front wing, suspension and sidepods, but low enough to keep the center of gravity down; streamlined fairings reduce the drag penalty.

C. Power, EMI, Cabling and Waterproofing

Compute modules, LiDARs and cameras draw significant power, so give them regulated DC–DC converters separate from the low-voltage ECU bus to avoid voltage sags. High-voltage harnesses and inverters produce strong electromagnetic interference (EMI), so use shielded twisted-pair cables for sensor data and route signal harnesses away from high-current cables. Clamp cables to the chassis to prevent chafing, and use connectors with a high ingress-protection rating (such as IP67) for wet running.

D. Thermal Management

Neural-network inference and point-cloud processing heat enclosed compute boxes. Use sealed fans or a conduction path to the chassis to prevent thermal throttling during long runs.

E. Trackside Calibration

A knock against a barrier can shift a bracket, so teams need a quick trackside routine using targets (checkerboards or reference poles) to re-verify the camera–LiDAR alignment between runs. Before each run the car should sit still on level ground so that gyroscope biases can be estimated and the INS can initialize.

F. Common Errors and How to Fix Them

Table V collects the most common sensor problems with their usual cause and fix. It is often the first place to look when something goes wrong at the track.

TABLE V. COMMON SENSOR PROBLEMS, CAUSES AND FIXES

Problem	Likely cause	Fix
IMU drift; heading wanders	Uncorrected bias, temperature change, no external reference	Warm up and estimate bias while static; fuse with GNSS and wheel speed; tune using Allan variance.
GNSS jumps or RTK fix lost	Multipath, obstruction, radio link loss	Better antenna placement (ground plane, clear sky); check the base-station link; fall back to IMU and odometry.
Magnetometer heading wrong	Motor, inverter and steel interference	Mount far from sources; hard/soft-iron calibration; use dual-antenna heading instead.
LiDAR ghost points	Dust, rain, direct sun	Range and intensity filters; clean the window; mount away from spray; tune ground removal.
Camera blinded or blurry	Glare, shadows, long exposure	Lens hood or polarizer; limit exposure time; HDR; fall back to LiDAR.
Cone color or position misaligned	Extrinsic shift, time offset	Recalibrate at the track; PTP/PPS time sync; rigid brackets.
Wheel-speed dropouts or noise	Inductive sensor at low speed, wrong air gap, EMI	Use Hall sensors; check the air gap; shielded cable; debounce in software.
A sensor stops publishing	Cable fault, USB disconnect	Screw-lock connectors; strain relief; watchdog and safe stop.
CAN data errors	High bus load, bad termination, noise	120 Ω termination at both ends; shielded twisted pair; reduce bus load; use CAN FD.

XI. SELECTION, COMPARISON AND TEAM HARDWARE

A. Sensor Comparison

Prices are rough planning estimates in Indian rupees (₹) and must be checked against current quotes. LiDAR range varies widely: low-cost solid-state units reach tens of meters, while mechanical units such as the Pandar40P reach about 200 m.

TABLE VI. APPROXIMATE COMPARISON OF SENSOR CLASSES

Sensor	Price (₹)	Range / accuracy	Power / mass
3D LiDAR	75,000 – 2,50,000	40–200 m; ±2 cm	10–20 W; 0.3–1.5 kg
Stereo camera	35,000 – 1,15,000	0.2–20 m; ±5 cm at 5 m	2.5–5 W; 150–350 g
Mono camera	15,000 – 45,000	Depth from AI or cone size	1.5–3 W; 80–180 g
RTK GNSS + INS	80,000 – 3,00,000	1 cm + 1 ppm (RTK)	3–8 W; 200–450 g
6-axis MEMS IMU	15,000 – 45,000	0.05 °/s gyro bias	0.5–1 W; 20–80 g
Hall wheel-speed sensor	1,500 – 8,000 per wheel	< 0.5 % error	< 0.5 W; 50–100 g

B. What Top Teams Use

Published system descriptions of leading FS driverless teams show a common pattern: one or more 3D LiDARs, one or more cameras (stereo and/or long-range mono), a GNSS/INS and, often, an optical ground-speed sensor for validation. Exact brands and mounting positions differ from team to team and from year to year, so each team's own publications should be cited when this section is finalized.

C. Budget, Mid-Range and High-End Configurations

Table VII sketches three tiers for a student team. Costs are indicative and include only the main sensors and the compute module.

TABLE VII. THREE SENSOR TIERS FOR A STUDENT TEAM

Tier	Sensors	Compute and cost
1. Budget	Livox Mid-360-class solid-state LiDAR; one global-shutter mono camera with YOLO; MEMS IMU (e.g., VN-100 class); four Hall wheel sensors	Jetson Orin Nano/NX; about ₹1,20,000–2,50,000
2. Mid-range	16–32-channel spinning LiDAR (Ouster OS1-32 class) or Livox Mid-360; ZED 2i-class stereo camera; GNSS-aided INS (VN-200 class) plus an RTK GNSS module	Jetson Orin NX or AGX Orin; about ₹2,50,000–5,00,000
3. High-end	Primary 32–64-channel LiDAR; two stereo or global-shutter cameras; dual-antenna RTK INS; optical ground-speed sensor; motor encoders	AGX Orin or industrial PC with mobile RTX; ₹5,00,000–12,00,000+

D. Our Hardware: History and Recommended Configuration

The current baseline is a Hesai Pandar40P LiDAR, a ZED 2i stereo camera, a VectorNav VN-200 GNSS-aided INS, Hall-effect wheel-speed sensors and a Jetson AGX Orin computer, with steer-by-wire and brake-by-wire actuation (Fig. 2). It is closest to Tier 3 in perception and compute, but with a single-antenna INS. All of these parts have vendor SDKs or drivers that work with ROS 2.

Recommendation for an Indian team. The aim is the best of both worlds: keep what already works and close the gaps that matter most, while choosing parts that can be bought,

serviced and replaced within India. Table VIII lists the baseline and the recommended configuration side by side.

TABLE VIII. BASELINE AND RECOMMENDED CONFIGURATION FOR AN INDIAN TEAM

Item	Baseline	Recommended	Reason
LiDAR	Pandar40P	Keep as primary; compare a Livox Mid-360 or 32-channel unit as a lighter, cheaper spare	Proven range; cheaper backup
Camera	ZED 2i	Keep; add one global-shutter camera if rolling-shutter blur appears	Fixes motion distortion
INS / GNSS	VN-200	Keep; add an RTK GNSS module (u-blox F9P class) and a local base station	Centimeter position, track mapping
Wheel speed	Hall sensors	Hall sensors with tone rings machined locally	Low cost; works at zero speed
Compute	AGX Orin	Keep; Orin NX as lighter spare	Headroom for neural networks
Actuation	Steer- and brake-by-wire	Dual position feedback, pressure sensors, independent EBS	Safety and rules
Time sync	–	PTP/PPS across LiDAR, camera and INS	Fusion alignment

Three reasons support this choice. First, keeping the existing LiDAR, camera, INS and computer avoids re-doing calibration, drivers and datasets. Second, the added items (RTK module, a spare camera, locally made tone rings) are inexpensive and easy to source, and they fix the weaknesses identified earlier: meter-level absolute position, possible rolling-shutter distortion and single points of failure. Third, a cheaper and lighter spare LiDAR gives the team a backup and lowers risk when imports are delayed (Section XII).

XII. SOURCING IN INDIA

Sourcing is often harder than selection: lead times, import duties and warranty support decide whether a sensor arrives before the competition. Prices and availability change often, so always request a current written quote.

A. Where to Buy

1) Authorized distributors and the manufacturer: For high-value items (LiDAR, INS, stereo cameras and Jetson modules), start with the manufacturer's list of distributors or resellers for India, and ask for an official quote, a warranty statement and the delivery terms. For example, NVIDIA Jetson modules are sold in India through authorized distributors, some of which advertise developer-kit support.

2) Indian robotics and electronics retailers: Hobby and robotics stores list many low-cost LiDARs, IMUs and Hall sensors. They are useful for prototypes and small parts, but most of their LiDARs are low-cost 2D or short-range units that are not suitable for FS cone detection.

3) Component distributors: Global component distributors that serve India (for example Mouser, DigiKey and element14) stock Hall-effect ICs, magnetic sensors, connectors, CAN transceivers and cables. They suit the parts needed for wheel-speed sensors and wiring harnesses.

4) Marketplace traders: Trade marketplaces such as IndiaMART list traders who sell Livox, Velodyne and Ouster LiDARs. They can be convenient but vary in reliability. Check that the unit is new, ask for the serial number and warranty terms, avoid paying in full before delivery, and cross-check with the manufacturer.

5) Direct import: Ordering directly from a manufacturer's online store is sometimes the only way to get a specific model; for example, the Stereolabs store lists ZED cameras with worldwide shipping. However this results in the buildup of additional costs such as customs duty, GST and shipping time.

6) Sponsorships and academic programs: Ask manufacturers and distributors about student-team discounts, sponsorship or loans, and check whether university research labs or other student projects can lend equipment.

B. Practical Tips

- Budget for import duty, GST and shipping. They can add a large share to the listed price and depend on how the product is classified.
- Order early. Lead times of weeks to months are normal for LiDAR and INS units, so build a spare-parts list (cables, connectors, mounts) at the same time.
- Confirm software support: driver and ROS 2 compatibility, SDK licensing and firmware versions.
- Prefer parts with datasheets, calibration certificates and a clear warranty, and keep invoices for customs and for scrutineering.
- Keep one cheaper backup sensor for each critical function (for example a low-cost camera or IMU) for testing and as a spare.
- Watch supply risk. The LiDAR industry is consolidating (Ouster acquired Velodyne in 2023, and Luminar filed for bankruptcy protection in late 2025), so check that a model is still in production and supported before standardizing on it.

XIII. TESTING, VALIDATION AND LESSONS LEARNED

A. Testing Individual Sensors

1) Static laboratory tests: Capture checkerboard or AprilTag targets across the whole camera frame to compute intrinsics and distortion. Keep an IMU still on a vibration-free surface for hours and run an *Allan variance* analysis to measure bias instability, drift and random-walk noise.

2) Controlled outdoor detection tests: Place blue, yellow and orange cones at known, tape-measured distances (for example every 2 m up to 30 m) and record camera detections, LiDAR points per cone and color accuracy at several angles.

Repeat in bright midday sun, low-angle glare and cloud shadow.

3) Dynamic localization tests: Measure RTK convergence time, signal recovery and drift while driving a surveyed loop. Run full-throttle launches, hard braking and slaloms to expose flexing mounts, resonance and connector dropouts.

B. Performance Metrics

Perception is measured by *precision* (correct detections out of all detections), which penalizes phantom cones:

$$\text{Precision} = TP / (TP + FP) \quad (4)$$

and by *recall* (correct detections out of all real cones), which penalizes missed cones:

$$\text{Recall} = TP / (TP + FN) \quad (5)$$

where TP, FP and FN are the numbers of true positives, false positives and false negatives. Color accuracy is the share of cones given the right color. Localization is scored with the mean absolute error (MAE) and the root mean square error (RMSE) between the estimated path and a ground-truth path (from RTK GNSS or an optical ground-speed sensor), together with lateral and yaw error. Latency is measured end to end (sensor exposure to actuator command) and per node (image inference, clustering, EKF update, planning).

C. Lessons Learned

- *Chassis flex shifts calibration.* Calibrating in a quiet workshop and then running on a flexing chassis misaligns camera and LiDAR. Use stiff, gusseted mounts and run trackside calibration checks before every session.
- *Unsynchronized clocks.* Fusing camera and LiDAR data without hardware timestamps (PTP or PPS) causes misalignment in fast corners. Synchronize every sensor to one master clock.
- *Over-reliance on simulated data.* Models trained only on clean simulator images fail on reflections, glare and unexpected backgrounds. To combat this one must record real-track ROSbags (A standard ROS 2 utility used to record and save all the data streaming through your nodes into a file) in varied weather to train and validate.
- *Silent sensor dropouts.* Nodes can freeze silently if a data frame is dropped or a cable disconnects, causing the system to run blindly without feedback (open-loop control). To prevent this, use physical locking connectors to secure your hardware, and implement strict ROS 2 watchdogs (monitoring timers) that automatically trigger a safe stop or the Emergency Braking System (EBS) if communication stalls.

XIV. Conclusion

This report walked through the sensors of a Formula Student driverless car, from basic specifications to the LiDAR, camera, GNSS/INS, inertial, wheel-speed and optical sensors, and the steer-by-wire and brake-by-wire actuation that closes the loop. The main lessons are practical: choose sensors whose limits match the track, synchronize and calibrate them carefully, fuse them so that one failure cannot blind the car, and mount and cable them as carefully as any other safety-critical part. For an Indian team, early ordering through authorized channels and a clear spare-parts plan matter as much as the sensor choice itself.

XV. Acknowledgment

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